Rossmann Sales Data Wrangling

2026-04-07

Data Wrangling

```
library(tidyverse)
library(lubridate)
```

Load the data

```
train <- read_csv("../data/raw/train.csv")
store <- read_csv("../data/raw/store.csv")
test  <- read_csv("../data/raw/test.csv")

head(train)
```

```
## # A tibble: 6 x 9
##   Store DayOfWeek Date       Sales Customers  Open Promo StateHoliday
##   <dbl>     <dbl> <date>     <dbl>     <dbl> <dbl> <dbl> <chr>
## 1     1         5 2015-07-31  5263       555     1     1 0
## 2     2         5 2015-07-31  6064       625     1     1 0
## 3     3         5 2015-07-31  8314       821     1     1 0
## 4     4         5 2015-07-31 13995      1498     1     1 0
## 5     5         5 2015-07-31  4822       559     1     1 0
## 6     6         5 2015-07-31  5651       589     1     1 0
## # i 1 more variable: SchoolHoliday <dbl>
```

```
head(store)
```

```
## # A tibble: 6 x 10
##   Store StoreType Assortment CompetitionDistance CompetitionOpenSinceMonth
##   <dbl> <chr>     <chr>                    <dbl>                     <dbl>
## 1     1 c         a                         1270                         9
## 2     2 a         a                          570                        11
## 3     3 a         a                        14130                        12
## 4     4 c         c                          620                         9
## 5     5 a         a                        29910                         4
## 6     6 a         a                          310                        12
## # i 5 more variables: CompetitionOpenSinceYear <dbl>, Promo2 <dbl>,
## #   Promo2SinceWeek <dbl>, Promo2SinceYear <dbl>, PromoInterval <chr>
```

Build Store Type A base tables

```
train_a_store <- train %>%
  left_join(store, by = "Store") %>%
  filter(StoreType == "a") %>%
  mutate(Date = as.Date(Date))
```

```r
test_a_store <- test %>%
  left_join(store, by = "Store") %>%
  filter(StoreType == "a") %>%
  mutate(Date = as.Date(Date))

glimpse(train_a_store)
```

```
## Rows: 551,627
## Columns: 18
## $ Store                     <dbl> 2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 14, 16, 17,~
## $ DayOfWeek                 <dbl> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5,~
## $ Date                      <date> 2015-07-31, 2015-07-31, 2015-07-31, 2015-07~
## $ Sales                     <dbl> 6064, 8314, 4822, 5651, 15344, 8492, 8565, 7~
## $ Customers                 <dbl> 625, 821, 559, 589, 1414, 833, 687, 681, 123~
## $ Open                      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ Promo                     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ StateHoliday              <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0",~
## $ SchoolHoliday             <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0,~
## $ StoreType                 <chr> "a", "a", "a", "a", "a", "a", "a", "a", "a",~
## $ Assortment                <chr> "a", "a", "a", "a", "c", "a", "c", "a", "c",~
## $ CompetitionDistance       <dbl> 570, 14130, 29910, 310, 24000, 7520, 2030, 3~
## $ CompetitionOpenSinceMonth <dbl> 11, 12, 4, 12, 4, 10, 8, 9, 11, NA, 3, NA, 1~
## $ CompetitionOpenSinceYear  <dbl> 2007, 2006, 2015, 2013, 2013, 2014, 2000, 20~
## $ Promo2                    <dbl> 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1,~
## $ Promo2SinceWeek           <dbl> 13, 14, NA, NA, NA, NA, NA, NA, 1, 13, 40, N~
## $ Promo2SinceYear           <dbl> 2010, 2011, NA, NA, NA, NA, NA, NA, 2012, 20~
## $ PromoInterval             <chr> "Jan,Apr,Jul,Oct", "Jan,Apr,Jul,Oct", NA, NA~
```

```r
glimpse(test_a_store)
```

```
## Rows: 22,128
## Columns: 17
## $ Id                        <dbl> 2, 3, 4, 5, 6, 7, 8, 10, 12, 13, 16, 18, 20,~
## $ Store                     <dbl> 3, 7, 8, 9, 10, 11, 12, 14, 16, 19, 22, 24, ~
## $ DayOfWeek                 <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4,~
## $ Date                      <date> 2015-09-17, 2015-09-17, 2015-09-17, 2015-09~
## $ Open                      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ Promo                     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ StateHoliday              <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0",~
## $ SchoolHoliday             <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ StoreType                 <chr> "a", "a", "a", "a", "a", "a", "a", "a", "a",~
## $ Assortment                <chr> "a", "c", "a", "c", "a", "c", "c", "a", "c",~
## $ CompetitionDistance       <dbl> 14130, 24000, 7520, 2030, 3160, 960, 1070, 1~
## $ CompetitionOpenSinceMonth <dbl> 12, 4, 10, 8, 9, 11, NA, 3, NA, NA, NA, 3, 1~
## $ CompetitionOpenSinceYear  <dbl> 2006, 2013, 2014, 2000, 2009, 2011, NA, 2014~
## $ Promo2                    <dbl> 1, 0, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1,~
## $ Promo2SinceWeek           <dbl> 14, NA, NA, NA, NA, 1, 13, 40, NA, 22, 22, 4~
## $ Promo2SinceYear           <dbl> 2011, NA, NA, NA, NA, 2012, 2010, 2011, NA, ~
## $ PromoInterval             <chr> "Jan,Apr,Jul,Oct", NA, NA, NA, NA, "Jan,Apr,~
```

Create train and test modeling panels for Store Type A

```r
train_a_model <- train_a_store %>%
  mutate(
```

```r
    Year = year(Date),
    Month = month(Date),
    DayOfWeek = wday(Date, week_start = 1),
    IsoWeek = isoweek(Date),
    IsWeekend = if_else(DayOfWeek >= 6, 1L, 0L),
    Open = coalesce(Open, 1L),
    Promo = coalesce(Promo, 0L),
    SchoolHoliday = coalesce(SchoolHoliday, 0L)
  )

test_a_model <- test_a_store %>%
  mutate(
    Year = year(Date),
    Month = month(Date),
    DayOfWeek = wday(Date, week_start = 1),
    IsoWeek = isoweek(Date),
    IsWeekend = if_else(DayOfWeek >= 6, 1L, 0L),
    Open = coalesce(Open, 1L),
    Promo = coalesce(Promo, 0L),
    SchoolHoliday = coalesce(SchoolHoliday, 0L)
  )

glimpse(train_a_model)
```

```
## Rows: 551,627
## Columns: 22
## $ Store                     <dbl> 2, 3, 5, 6, 7, 8, 9, 10, 11, 12, 14, 16, 17,~
## $ DayOfWeek                 <dbl> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5,~
## $ Date                      <date> 2015-07-31, 2015-07-31, 2015-07-31, 2015-07~
## $ Sales                     <dbl> 6064, 8314, 4822, 5651, 15344, 8492, 8565, 7~
## $ Customers                 <dbl> 625, 821, 559, 589, 1414, 833, 687, 681, 123~
## $ Open                      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ Promo                     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ StateHoliday              <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0",~
## $ SchoolHoliday             <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0,~
## $ StoreType                 <chr> "a", "a", "a", "a", "a", "a", "a", "a", "a",~
## $ Assortment                <chr> "a", "a", "a", "a", "c", "a", "c", "a", "c",~
## $ CompetitionDistance       <dbl> 570, 14130, 29910, 310, 24000, 7520, 2030, 3~
## $ CompetitionOpenSinceMonth <dbl> 11, 12, 4, 12, 4, 10, 8, 9, 11, NA, 3, NA, 1~
## $ CompetitionOpenSinceYear  <dbl> 2007, 2006, 2015, 2013, 2013, 2014, 2000, 20~
## $ Promo2                    <dbl> 1, 1, 0, 0, 0, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1,~
## $ Promo2SinceWeek           <dbl> 13, 14, NA, NA, NA, NA, NA, NA, 1, 13, 40, N~
## $ Promo2SinceYear           <dbl> 2010, 2011, NA, NA, NA, NA, NA, NA, 2012, 20~
## $ PromoInterval             <chr> "Jan,Apr,Jul,Oct", "Jan,Apr,Jul,Oct", NA, NA~
## $ Year                      <dbl> 2015, 2015, 2015, 2015, 2015, 2015, 2015, 20~
## $ Month                     <dbl> 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7, 7,~
## $ IsoWeek                   <dbl> 31, 31, 31, 31, 31, 31, 31, 31, 31, 31, 31, ~
## $ IsWeekend                 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
```

```r
glimpse(test_a_model)
```

```
## Rows: 22,128
## Columns: 21
## $ Id                        <dbl> 2, 3, 4, 5, 6, 7, 8, 10, 12, 13, 16, 18, 20,~
```

```
## $ Store                     <dbl> 3, 7, 8, 9, 10, 11, 12, 14, 16, 19, 22, 24, ~
## $ DayOfWeek                 <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4,~
## $ Date                      <date> 2015-09-17, 2015-09-17, 2015-09-17, 2015-09~
## $ Open                      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ Promo                     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ StateHoliday              <chr> "0", "0", "0", "0", "0", "0", "0", "0", "0",~
## $ SchoolHoliday             <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ StoreType                 <chr> "a", "a", "a", "a", "a", "a", "a", "a", "a",~
## $ Assortment                <chr> "a", "c", "a", "c", "a", "c", "c", "a", "c",~
## $ CompetitionDistance       <dbl> 14130, 24000, 7520, 2030, 3160, 960, 1070, 1~
## $ CompetitionOpenSinceMonth <dbl> 12, 4, 10, 8, 9, 11, NA, 3, NA, NA, NA, 3, 1~
## $ CompetitionOpenSinceYear  <dbl> 2006, 2013, 2014, 2000, 2009, 2011, NA, 2014~
## $ Promo2                    <dbl> 1, 0, 0, 0, 0, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1,~
## $ Promo2SinceWeek           <dbl> 14, NA, NA, NA, NA, 1, 13, 40, NA, 22, 22, 4~
## $ Promo2SinceYear           <dbl> 2011, NA, NA, NA, NA, 2012, 2010, 2011, NA, ~
## $ PromoInterval             <chr> "Jan,Apr,Jul,Oct", NA, NA, NA, NA, "Jan,Apr,~
## $ Year                      <dbl> 2015, 2015, 2015, 2015, 2015, 2015, 2015, 20~
## $ Month                     <dbl> 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9, 9,~
## $ IsoWeek                   <dbl> 38, 38, 38, 38, 38, 38, 38, 38, 38, 38, 38, ~
## $ IsWeekend                 <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
```

In addition to sales, several time-varying explanatory variables were included. The Promo variable represents the proportion of stores running promotions on a given day, while SchoolHoliday and StateHoliday capture the proportion of stores affected by school and public holidays, respectively. These variables provide additional context for changes in sales beyond purely temporal patterns.

Aggregate Store Type A to complete daily train and test series

```r
train_a_agg_daily <- train_a_model %>%
  group_by(Date) %>%
  summarise(
    Sales = sum(Sales, na.rm = TRUE),
    OpenStores = sum(Open == 1, na.rm = TRUE),
    PromoStores = sum(Promo == 1, na.rm = TRUE),
    SchoolHolidayShare = mean(SchoolHoliday, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    PromoShare = if_else(OpenStores > 0, PromoStores / OpenStores, 0)
  ) %>%
  complete(
    Date = seq(min(Date), max(Date), by = "day"),
    fill = list(
      Sales = 0,
      OpenStores = 0,
      PromoStores = 0,
      PromoShare = 0,
      SchoolHolidayShare = 0
    )
  ) %>%
  mutate(
    SchoolHolidayShare = replace_na(SchoolHolidayShare, 0),
    Year = year(Date),
    Month = month(Date),
```

```r
    DayOfWeek = wday(Date, week_start = 1),
    IsoWeek = isoweek(Date),
    IsWeekend = if_else(DayOfWeek >= 6, 1L, 0L)
  ) %>%
  arrange(Date) %>%
  select(
    Date,
    Sales,
    OpenStores,
    PromoStores,
    PromoShare,
    SchoolHolidayShare,
    DayOfWeek,
    Year,
    Month,
    IsoWeek,
    IsWeekend
  )

test_a_agg_daily <- test_a_model %>%
  group_by(Date) %>%
  summarise(
    OpenStores = sum(Open == 1, na.rm = TRUE),
    PromoStores = sum(Promo == 1, na.rm = TRUE),
    SchoolHolidayShare = mean(SchoolHoliday, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    PromoShare = if_else(OpenStores > 0, PromoStores / OpenStores, 0)
  ) %>%
  complete(
    Date = seq(min(Date), max(Date), by = "day"),
    fill = list(
      OpenStores = 0,
      PromoStores = 0,
      PromoShare = 0,
      SchoolHolidayShare = 0
    )
  ) %>%
  mutate(
    SchoolHolidayShare = replace_na(SchoolHolidayShare, 0),
    Year = year(Date),
    Month = month(Date),
    DayOfWeek = wday(Date, week_start = 1),
    IsoWeek = isoweek(Date),
    IsWeekend = if_else(DayOfWeek >= 6, 1L, 0L)
  ) %>%
  arrange(Date) %>%
  select(
    Date,
    OpenStores,
    PromoStores,
    PromoShare,
```

```r
    SchoolHolidayShare,
    DayOfWeek,
    Year,
    Month,
    IsoWeek,
    IsWeekend
  )

head(train_a_agg_daily)
```

```
## # A tibble: 6 x 11
##   Date         Sales OpenStores PromoStores PromoShare SchoolHolidayShare
##   <date>       <dbl>      <int>       <int>      <dbl>              <dbl>
## 1 2013-01-01    2907          1           0          0             1
## 2 2013-01-02 3920386        602           0          0             1
## 3 2013-01-03 3528027        601           0          0             0.939
## 4 2013-01-04 3610215        601           0          0             0.939
## 5 2013-01-05 2964587        600           0          0             0.0664
## 6 2013-01-06    6254          3           0          0             0.0664
## # i 5 more variables: DayOfWeek <dbl>, Year <dbl>, Month <dbl>, IsoWeek <dbl>,
## #   IsWeekend <int>
```

```r
head(test_a_agg_daily)
```

```
## # A tibble: 6 x 10
##   Date       OpenStores PromoStores PromoShare SchoolHolidayShare DayOfWeek
##   <date>          <int>       <int>      <dbl>              <dbl>     <dbl>
## 1 2015-08-01        461           0          0              0.113         6
## 2 2015-08-02         12           0          0              0.113         7
## 3 2015-08-03        461         461          1              1             1
## 4 2015-08-04        461         461          1              1             2
## 5 2015-08-05        461         461          1              1             3
## 6 2015-08-06        461         461          1              1             4
## # i 4 more variables: Year <dbl>, Month <dbl>, IsoWeek <dbl>, IsWeekend <int>
```

Additional outputs (Version 1 and Version 2)

```r
# Version 1: clean univariate daily sales series
df_A_daily_clean <- train_a_model %>%
  group_by(Date) %>%
  summarise(
    Sales = sum(Sales, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(Date) %>%
  mutate(
    Year = year(Date),
    Month = month(Date),
    Week = week(Date)
  )

# Version 2: daily series with external variables
df_A_daily_full <- train_a_model %>%
  group_by(Date) %>%
```

```r
  summarise(
    Sales = sum(Sales, na.rm = TRUE),
    n_stores = n(),
    Promo = mean(Promo, na.rm = TRUE),
    SchoolHoliday = mean(SchoolHoliday, na.rm = TRUE),
    StateHoliday = mean(if_else(is.na(StateHoliday) | StateHoliday == "0", 0, 1), na.rm = TRUE),
    .groups = "drop"
  ) %>%
  arrange(Date) %>%
  mutate(
    Year = year(Date),
    Month = month(Date),
    Week = week(Date)
  )

head(df_A_daily_clean)
```

```
## # A tibble: 6 x 5
##   Date         Sales  Year Month  Week
##   <date>       <dbl> <dbl> <dbl> <dbl>
## 1 2013-01-01    2907  2013     1     1
## 2 2013-01-02 3920386  2013     1     1
## 3 2013-01-03 3528027  2013     1     1
## 4 2013-01-04 3610215  2013     1     1
## 5 2013-01-05 2964587  2013     1     1
## 6 2013-01-06    6254  2013     1     1
```

```r
head(df_A_daily_full)
```

```
## # A tibble: 6 x 9
##   Date         Sales n_stores Promo SchoolHoliday StateHoliday  Year Month  Week
##   <date>       <dbl>    <int> <dbl>         <dbl>        <dbl> <dbl> <dbl> <dbl>
## 1 2013-01-01    2907      601     0        1             1      2013     1     1
## 2 2013-01-02 3920386      602     0        1             0      2013     1     1
## 3 2013-01-03 3528027      602     0        0.939         0      2013     1     1
## 4 2013-01-04 3610215      602     0        0.939         0      2013     1     1
## 5 2013-01-05 2964587      602     0        0.0664        0      2013     1     1
## 6 2013-01-06    6254      602     0        0.0664        0.244  2013     1     1
```

Sanity checks

```r
date_span <- as.integer(max(train_a_agg_daily$Date) - min(train_a_agg_daily$Date)) + 1
missing_days <- date_span - nrow(train_a_agg_daily)

tibble(
  n_store_type_a_train_stores = n_distinct(train_a_model$Store),
  n_store_type_a_test_stores = n_distinct(test_a_model$Store),
  n_train_rows = nrow(train_a_model),
  n_test_rows = nrow(test_a_model),
  n_train_agg_daily_rows = nrow(train_a_agg_daily),
  n_test_agg_daily_rows = nrow(test_a_agg_daily),
  missing_days_in_agg = missing_days
)
```

```
## # A tibble: 1 x 7
```

```
##   n_store_type_a_train_stores n_store_type_a_test_sto~1 n_train_rows n_test_rows
##                         <int>                     <int>        <int>       <int>
## 1                         602                       461       551627       22128
## # i abbreviated name: 1: n_store_type_a_test_stores
## # i 3 more variables: n_train_agg_daily_rows <int>,
## #   n_test_agg_daily_rows <int>, missing_days_in_agg <dbl>
```

Save processed outputs

```r
dir.create("../data/processed", showWarnings = FALSE, recursive = TRUE)

write_csv(train_a_model, "../data/processed/store_type_a_train_store_daily.csv")
write_csv(test_a_model, "../data/processed/store_type_a_test_store_daily.csv")
write_csv(train_a_agg_daily, "../data/processed/store_type_a_train_aggregate_daily.csv")
write_csv(test_a_agg_daily, "../data/processed/store_type_a_test_aggregate_daily.csv")
write_csv(df_A_daily_clean, "../data/processed/daily_store_a_clean.csv")
write_csv(df_A_daily_full, "../data/processed/daily_store_a_full.csv")
```